

ActInf Textbook Group ~ Cohort 8 ~ Session 3

(Chapter 2) ~ 3/6/2025

Source: [YouTube](#) **Playlist:** Active Inference Textbook Group ~ Parr et al. 2022 ~ Cohort 8 **Duration:** 1:12:53 | **Uploaded:** Unknown **Transcript method:** auto_caption

okay I believe it is recording so welcome everyone we're in our first meeting for cohort 8 in Chapter 2 so we'll head over to the chapter 2 page and to chapter 2 in the book and let's just begin does anyone have any thoughts quotes sections that they'd like to pick up with about chapter 2 like what was something that stayed with them about chapter 2 whether just a phrase or what is the low road what did you expect the low road to be what did you walk away feeling that the low road [Music] was I I like the question I just had so did you tell me when that's most relevant place to put that question in the audience maybe ask again just to be clear so I feel there are two levels of looking at the text uh when we talk about the action perception cycle and uh it says that we have preferences but it can be interpreted I think in two ways the body having preferences or me as the experiencing subject having those preferences and so it feels like it alternates between those Cycles as if it's the same thing at least it does not distinguish quite clearly between those two levels of looking at it uh and I was thinking is this distinction even analytically Justified and second if so how then should one this alternate between those two lenses obviously there's a lot of ways to go into that just to respond narrowly and analytically active inference does not require a nested or a multi-level model the first part of the book is going to build up towards just looking at single layer models so not going into higher order human type cognition differentiating between minds and bodies and aware and subconscious and multiple types of experiences and all of this it doesn't have to be an embodied system we could just be talking about a dynamical oscillating system or even simpler systems where bringing out this framework would be total Overkill so no specific type of articulation between any kind of body and mind is presupposed by the framework at all which is why it almost always comes back to active inference saying less so you can say more so the inquiry starts to look like you posit a kind of structure for a given body and mind and experiencer and then what kinds of experiments or measurements reduce your uncertainty

about that space of hypothesis but it's not to be found in the framework I I do acknowledge the the general class of subject matter of active influence uh but they do also claim that we as humans fall under the subject matter we are a cell systems and therefore we are also uh subjected to this notion of having these pre preferences like also statical dynamical systems could have those preferences um but then you know preferences seem to have a double layer we experience ourselves as having preferences and most of the times we are like conscious of it and those preferences ascribed To Us by the active inflence Paradigm and then how do these two Notions relate to each other so so it sounds like you're you're trying to distinguish between two sets of preferences is that right uh the preferences uh we should have uh by the active inference framework which sees as systems and we have preferences like those are like the Tendencies we we see my systems to to evolve towards and to also avoid entropy those kind of preferences ascribed by the active influence framework but also the active preferences we as normal humans seem to experience that we have a certain will and certain desire and a wish to go to certain States for and and those things seem to contradict at first glance for me you know with depression or like the Suicidal Tendencies etc etc so how do you feel about this so uh the appeal to me so far has been that I think those personal preferences are also dictated by these sort of like entropic forces that um steer us towards I don't know um towards like a a set of preferences that is uh I don't know how explain it yet but just to throw it in because I also thought about certain kind of religious as such but someone can for example smoke having full Consciousness about the harmful effects about his losing his life span destroying his lungs and yet do the act of smoking so how does do two things relate then you think so this is I think this is like the um the one of the most interesting Parts about this I think so in in some way somewhere along the line that person has been I guess building a set of preferences that um um makes them probably more comfortable to you know choose smoking or something there there's some reason that this person has to be choosing these things because maybe it makes them feel a certain way about themselves or about how they fit in the world or something um you know it's probably so okay I think I think I understand what the general theme is about active inference explanations and is that if this seems to be discrepancies between those two level preferences at the system base and experimental base then explanations to be s in the complexity of the model which seemingly can for example can contradict the lower the lowest simplistic systems of the model so that the answer is always then to be s in the complexity of the system is that you think the right how else I think Malcolm's trying to talk but I can't really hear him go ahead Malcolm okay volume thing I mean sure short answer yes nested models or

interacting models of this is true of a linear aggression model this is just general to any kind of explanation of variance if something is not being explained by a model structure within a level often it's it's um accounted for elsewhere but like before jumping to the psychological connections of the terms like preference and trying to associate it like you gave an example of a behavior so even just speaking really broadly with a single layer Behavior so it could be that the Habit the E the prior on action is such that a given action is selected like turning a certain direction or you could localize the repetitive behavior causality in the be so the person has they don't understand the consequences of a given action like their skin itches and they don't know that not that doing this will do that or it could be Inc in preference it's like so the structure of the inquiry is to develop these kinds of precision Psychiatry or cognitive models to differentiate hypothesis so thinking about it qualitatively speculatively mixing ontologies combining experiential and Technical elements those are early and sort of generative phases and then yes you start to specify generative models following like the chapter six type recipe or using other approaches and then you start to look at the relative merits of these different approaches let let's return to yeah yeah okay M go for it oh you're muted but then continue yeah I don't want to straight the topic too much on one end and if someone else supposed to to speak as well yeah let's please return to chapter two and if somebody has a specific quote or or question or idea about chapter 2 I just wanted to to add one more idea I I had up your um question mon like the idea of contradicting preferences and how system would like as you said maybe you want to smoke like you have this preference of wanting to smoke and wanting to smoke and how this affects your um selection of a policy of an action and actually I was just reading yesterday a nice paper of like the attachment style of where this experiment where the children are playing then the mother gets out and they are crying then she gets back in and depending on the attachment style the the curely attached child would will be angry but still get to the mom and knows what to do while the um disorganized child will like be in a unsettled mood when the mother gets in and it's not really like a clear policy so there's a high uncertainty like when you um refer to the active inference um selection policy selection calculation a high uncertainty what policy is actually the best policy I just should I should choose in that moment given the observations I have and the contradicting preferences something great Ben's liing a little bit Yeah the example you offered with the smoking will maybe Behavior as we maybe see quit smoking but I don't know thank you Ben yeah just a short comment yeah I I do acknowledge like there's a lot of explanatory power in active influence yes okay awesome so let's kind of return to a chapter two anyone you know write in

the chat or or yeah um so I okay I can start with something in chapter two sure um so one of the sort of questions I was trying to follow along in the chapter to see if there was an answer for is um is uh essentially like how how a decision is actually made so they talk a lot about a couple times about how there's like a balance between these two terms in the um variation of free energy right how one of them is like you know is about exploring the space and one of them is about um exploiting the space um but I wasn't really clear on how a decision is actually made so they talk about how like um the policy is like the relative balance between these two terms but at what point is a decision made is it like is there a threshold somewhere that um like an agent actually uses to say okay actually I need more information so I'm going to explore the space more versus like okay I actually I want to obtain this this preferred State more than I care about getting information so I'm going to um you know uh you know seek out this state or something it like I guess my question is is there a threshold like how is a decision actually made yeah Sylvia do you want to add something to that yes and I feel like maybe that threshold could be determined by um that agents prior or preferences to begin with for example even personality some people tend to be more Risk conservative While others are Risk Takers and in a way that fresh would differ from those people or we look at people with psychiatric disorders they have very rigid pray about the reality that is clearly kind of Mal adductive even but to them it makes sense to choose certain policies or make decisions at a certain point while normal people might think that well that's not rational or that doesn't even make sense so I guess the fres home might be very dependent on the contest and also that agents like prior to begin with for that specific like goal or desirable outcome that they are pursuing after okay so so there is a threshold then that is the idea um yes I I believe so another another aspect of this just operationally when you implement these models is here's you know one one tutorial one place to look at this is you have a prior on policy so that's before the observation comes in How likely are you to do different things 50% to turn left 50% to turn right an observation comes in the policy prior is updated to the policy posterior through the calculation of expected free energy so the prior on policy was even and then the pragmatic value and the epistemic value of these two Alternatives reweight them so now it's 70% likely to turn right 30% to turn left and then from that posterior distribution one action is sampled and then that is emitted so in this simulation every time Step In Action is decided now the question of when an action is taken is done with a few different approaches one approach is to have a null action like have weight as an action another approach is to have a nested model where at the highest level of the model you have weight or act and then within act you have

just the choices of action gotcha okay an action is selected because there's an unrolling continuous action so then it's just are you trying to model something that happens at every single time step like gas or break but then if you want to include neither you could either include that in the base level or you have this nested model with and then and then that's where would you'd get into variation in those cognitive parameters what are someone's priors on How likely they are to use the gas or the BR what are the B Matrix reflecting people's understanding of what happens when you hit the gas or the break what are their preferences about what they expect to see and prefer to see from sensory observations achieved by hitting the gas in the break got it that's that makes sense so it's like yeah it's it's a sampling of policies not just a simple threshold It's s actions over policies one action gets sampled so carried through time is a distribution over action choices and then that gets sampled one I mean this is a great line and this is like when it's like wait what is active inference this is a an important thing to walk away from chapter 2 with what is the simple but fundamental Advance offered by active inference this is the question so that's that's an interesting thing to to ask and learn and how does that fit into the low road okay let's look at some of the questions in chat quickly okay John would you say preferences are revealed by priors which are revealed by observations I mean revealed to whom when you're the one writing the generative model you wrote all those variables in your code in a given situation where you're reducing your uncertainty about those parameters you could imagine a situation where you're reducing uncertainty about one of those variables based upon your assessment of another one but sure there could be a situation where through observation which is kind of the ultimate empirical interface you are learning about priors but priors are a bit broad because priors in Bas in statistics refer to basically anything before they get updated but then they become the posterior relative to them being prior to that observation but the posterior is again prior to the next observation so it's usually a little bit more helpful to specify like prior on what um can I answer that uh also to John so yeah I was thinking about this a couple days ago this is really interesting because what do you have when some like say somebody experienced something new for the first time maybe they've I don't know they've never gone like rock climbing or something and they were seen it before and then they go and all of a sudden it's really exciting and they're they have all these different experiences about rock climbing or something and now they love it and they didn't know that before so in that case their their their priors were revealed by observations right like there has to be some mechanism in this where somebody can like you know be adding onto their their priors because there has to be new you know

room for new experiences to be had so in to that question I think I would say yes yes Al great point also within the textbook like within the early part of working within a single layer model and so on it's a structurally fixed model so the thermometer is dealing in degrees temperature and it can quantitatively adjust within degrees temperature but it's not going to start seeing or modifying color that gets into the space of structure learning where you have model structure as an inference Target and that's where the Assembly of new Concepts and changes in the structure of affordances or or anything come into play um okay Andreas what's the difference between predictions and preferences so as used here preferences are a prior Distribution on observation that reflect expectations over observation consistent with being a cost or a constraint function also sometimes called preferences and it's used in pragmatic value calculation predictions are anticipations about something usually about observations okay anyone again can can write something or or um raise their hand but gu you could write that down somewhere it's all recorded you can return to but yeah and and also the these definitions are the active inference ontology the definitions of terms and and just it's just about understanding um where is it referring to like continent scale like prior again but it could be a prior on anything so there's little that can be can't go too much into saying what about this prior because it's just like a prior on what so a lot of it is just knowing when do you need to just add more detail and when it comes down to it it's about the specific generative model constructed um but thinking about the structure of the book a little bit remember we're in the first half so we're in the background and theory and learning area chapter 1 was introductory chapter 2 and three are going to bring us to active inference from two different directions the low road and the high road chapter 2 is going to do this from the low road chapter 3 is going to do this from The High Road chapter 4 is going to be the active inference generative model which is kind of like this boundary artifact which is the target of our construction it's used in computational inference looked at from the low road it's constructed According to some of these topics looked at from the high road it's consistent under the free energy principle chapter four chapter five is a review on a few different generative models that have been constructed in Maman human Neuroscience so that's the first half of the book is Introduction and then two roads converging on a certain type of artifact that we construct chapter 5 giving examples of how that artifact is used so it's it's um it's it's going to be like two different complimentary approaches the low road is like a build it from the ground up computer science or engineering starting with probabilistic modeling and Baye equation working towards building integrated models that have both perceptual inference and planning or action selection as inference that's

like the build it from the ground up it's like if you were building a cognitive framework for an agent embodied or cyber physical or whatever it was and you wanted to combine the inbound signal processing element with the outbound action selection control theoretic element and you wanted to integrate the inbound and the outbound into a unified model and then apply it to things like brains using tractable approximations like variational inference that's kind of what happens on the low road is like building cognitive models that unify perception cognitive phenomena and action in a ground up way we'll get to chapter 3 in the high road when we get there but that's sort of the low road and and that's um how from just primary considerations in statistical modeling of cognitive systems one reaches something basically like active inference just from considering the question of modeling agents that's in figure 1.2 but then chapter 2 goes into this in more depth and starts with this again kind of Step by Step bottom up but for those who haven't seen much of Bayesian Statistics before this is kind of a good Motif like we have a prior belief about objects in our hand whether they're frog or apple not a secret third thing again that would be the structure learning um and then an observation comes in and the beliefs about the object are updated that's perceptual inference and something analogous occurs with action inference where we have a prior on what we are apt to do then the observation comes in and then we have a posterior on what we're apt to do which get sampled from and that was kind of the question earlier okay anyone want to ask something or I can copy one of the questions from the chat and yeah okay Henry yeah hi so I have a I have a question about equation equation 2.5 so the variational free energy definition yes um so I guess these are two quite simple but interrelated questions the first is like um in the complexity versus accuracy factorization um oh sorry the Divergence the Divergence in evidence factorization the uh the Divergence is like computed with respect to the approx posterior um when like canonically the you'd compute this with respect to the uh the true the true distribution is this just because if it was the other way around it wouldn't be equivalent to like the energy entropy formation and it's a like a post talk justification or or is it per more like perers than than that um does that make sense can we talk talk freely or sorry is it like uh directed questioning or can I just respond to this question if you have a thought yeah yeah yeah um yeah so my my understanding there was that it was with respect to the variational because we don't know the true posterior right so that I mean that's the whole point right like we use Q because we don't know what p is so we approximate it with Q uh yeah I I was just wondering because like that that final term like the um the additional quantity over the the true evidence like we can't we can't compute that anyway because we we can't compute the the true

the true posterior so we'd use one of the other two factorizations to like plug in plug in the numbers and and we show their equivalent to the third is that is that correct yeah two two um answers here just practically one of them is when you implement the equation these three decompositions are equivalent but when you look at the code like for pmdp or something like that you will find it's often only implemented using one of them and there's probably situations where implementing using one of these decompositions or another have different computational performance a second answer that's really important is you pointed out that the evidence term is fixed and doesn't depend on Q so when we're interested in variational gradient descent the evidence term is actually a way to kind of quarantine the overall likelihood of the data and only follow ΔF which is ultimately what we're going to use anyway that's why the overall actual F will be like 83 1114 but that's that's like looking at the least squares error for a linear aggression model it doesn't matter what the real number is it's about within a model like following the gradient on it so this is a really good way just to say well whatever the evidence is whatever the likelihood is of the data we're going to minimize the Divergence by doing so and following that Tailwind minimize the variational free energy whether that convex optimization happens to be up here or down here um Sylvia um yeah I have a more General question but just adding to this I was thinking when we talk about um free energy or variation of free energy and saying that it has an upper bound is it correct to actually perceive it or simplify it as like you know to see that it could be an overestimate of the actual surprise the surprise that we can't really you know no the the true uncertainty that is out there but we are giving it an estimate to make sure that we can reduce it regardless you know by our model like is that a correct way of perceiving it yeah absolutely so when the process generating observations is the same model family as the variational distribution so gaussian process in the world and we're going to use a gaussian variational estimator then the variational free energy will converge to the actual generative function whereas if the generative function in the world were something just weird or or bodal you still within the variational family of choice would be doing the best you can by reducing the Divergence but that could be hopelessly far away so that's why the it's an upper bound that's always equal to or greater than because the best you could do would be if the variational family were the right family to exactly become congruous with the data generating function but in the situation where the data generating function is anything else then you'll be minimizing your variational free energy to some point slightly um not exactly minimizing surprise on the target distribution but it's the best that could be done within that constrained easily optimizable

interpretable family yes that makes sense um the general question that I had was um as we were talking about advancement and what are the key implications of active influence treating perception and action as part of the same cycle with this shared objective rather than the separate processes with like distinct goals and how does this unified framework this coherence apply to real world I'm just really curious about how everyone sees the potential of at influence in relation to like advancing their FS because like for me with someone having a background in Psychology I think it's a really nice way of putting very fragmented theories in different branches together and to have like one unified um explanation so I'm just curious about you know how other people are taking it um from the perspective of their fields anyone can go for it but those are awesome questions that's like literally the whole Fields but are we gonna have a special theory for every brain region every cognitive phenomena or what happens when we do have unified approaches yeah M than anyone else yes I was just AI like wondering because I do not know this the scope of active inference explanatory power so also know it's limits so I was actually interested are there any like limits that what we can speak of about till we can only use inference for these matters but not for these matters is there anything or do you guys if you Dani fman I think you're most red well red on active inference uh think it's everything that they can explain anything could be modeled but whether to anyone any given model of a given thing is satisfactory is a totally outside question so the explanatory scope of the framework generically is similar to like saying well what's the explanatory scope of Statistics you could model any distribution but does it say anything about the height of these children in this classroom or why this happens here no but could it be used to model why that happens there for that thing yes is that a tiny engineering project or is that a mega engineering project that's the question is it something where we're going to reach a satisfying account easily uh um the purpose I I can I forget who's the name of the psychologist the lady psych anyway my own purpose is trying to figure out how to integrate or find coherence across uh Dimensions that are operating when you watch a movie or a video basically the the text or the copy the uh the cinematography and the music they're all coming in through different perceptual systems but they have to create unified immersive experience and the predictions we're trying to make uh our audience responses I've actually done some experiments with that because we have ways of separating the cinematography from the storytelling from the the meaning being communicated by the music and how that parallels the meaning being communicated by the uh um uh in terms of semantics and so on so that you know you know that's that just just is a way of saying how I think this stuff will apply I'm

trying to predict audience responses to creative and it turns out AI is really good at coding the creative and making guesses already about what's going on and then when I show it actual audience responses like oh I was wrong here I was wrong there and so I'm trying to improve the model over time and that's that's actually why I'm in the class awesome that's going to come up like in chapter six this is there's some latent underlying phenomena like a frog or a well performed multimodal performance well actually an interesting thing we just ran across the other day I don't know if you ever heard of tsloot subjective correlative no it's it's a literary theory that the you know the Nobel prize winning poet TS earlier came up with basically saying that if you want to evoke a specific emotion there is there is a unique sequence of objects in your perceptual world that if you go through that sequence will evoke that very particular emotion it says so that there's actually an objective sequence of perceptions that lines up with your associative memory system basically I you wrote an essay with it and stuff I'll try to forward it to the group cool I mean this is kind of a general madlib structure is re-approaching X from an active inference perspective like what was pragmatism about or what was this about or how could this be yeah those are especially if those connections haven't been made writing those pieces out and footnoting it is an awesome way to to open up your thinking and footnote it and also well if you take it one step further it's actually a way of generating new data that too you know so that you know so the goal is obviously get to a self-learning situ situation every time you connect ideas because you're making connections that the AI isn't it just it's a Continuum it knows a lot of stuff but if you connect the dots for them you're like being a neuron in the thing basically it can it it can create new do data on on meaningful connections that it wouldn't see without your help I think that you know driving towards a central role for human beings in the interaction with Ai and and and and and and the the theoretical framework being something like active so okay um mon then AI I was also very interested in thinking about how uh to interpret what active INF is doing if you're like a modeler in mind so uh so there's like some higher goal to be happy for example and to be happy you have found a sub goal method for that and that in itself also becomes a goal and then you ask for the same goal how to achieve this etc etc then you guy kind of see this this scheming down way and how how does this thinking relate to active inference and how they choose certain policies like a sequence of actions which generates observations in line with some prior um is this this this notion of high will low will to be see in the nested models or uh should we like relieve all these Notions of old and only think of active terms lot of ways to go with that like keep that feeling and channel as open as it can be and really look at the textbook

for what it says and then whether those Lego pieces were exactly what you sought or whether you see that you need to build up a ton of pieces to get to the sophistication and the interplays that you're talking about then you'll have the grammar to go about that but you may find that it's literally addressed within a single layer and that you're talking about the interplay of two kinds of variables within a simple model at a single layer or you may find that you're thinking more about nested models and then you'll have the grammar to actually go about building those Adam uh yeah I I was so regarding equations 2.5 and I think 2.6 Etc I was wondering if those are derived anywhere mathematically and because it seems like they're derived sort of theistically is there anywhere where I can see where they're actually yeah yeah so um here in the equation rows um um here and these are just tiny examples of the absolutely transformative contributions that people can make first off with their written questions also with derivations if they understand so this is I believe from yakob smackle here is equation 2.5 oh you know these are materials that aren't in the textbook The Textbook is just like the tip of the iceberg and we can go a lot further and connect a lot of the paths so here's 2.5 one of the big pieces about 2.5 that that um like really helps clarify it is the the route of derivation is actually Row one goes to two and Row one goes to three there may also be a path to go from two to three but that was very helpful to understand that there's a path from 1 to two and 1 to three um there should be if they're equivalent right yeah there better be yeah yeah I'm sure there's a way to do it but just in terms of ones that can fit on a single slides yeah and then 2.6 um Jonathan shock has let's see if this link still works there there may be another updated or published version but Jonathan shock who's a a math professor um yeah he's done some derivations on expected free energy as well okay nice yeah so those are good um yeah here's his and this is in the math learning group so like you know it's awesome that people have all these questions like that's literally what the questions are for is is just add a question and write it out everything that people are typing and here adding the resources that are relevant like please in the time between enrich what we have and and add more and add your thoughts like imagine that these questions are being asked by people right now and the ones that you're excited about adding information to just go for it okay nice yeah there's a ton of material on the on this we I haven't really gotten to half of it yet so yeah this is also is kind of a fun piece about applying active inference to topics like this was just done with all by all I generated synthetic domains and then I just applied each domain to and from active inference using IIm so this got ridiculously far oh in starting to to apply active inference to you know ATM transactions and blockchain and chemistry and things like that so understanding how to even

use the exence tools to project to and from active inference ontology and techniques is already extremely productive okay very cool thanks thanks for that yeah was someone else going to say something I have I have one more quick go for it um so on page 37 figure 2.6 I kind of wanted to understand that diagram just a little bit I think I do so um my understanding of this is saying okay so like at the top level there's the equation for the for the expected free energy and then below it are different decompositions you can make if you omit certain terms so like would that cor or does that correspond to like if you um if you know you have a different I don't know like decision regime in your certain use case would you decide to Omit a certain term because you're trying to um I don't know like structure the learning that way or something or like how do you know when you want to admit a certain term or Etc yeah great way to put it so this is like equation 2.6 um where we have the expected free energy here usually people talk about this first decomposition between the epistemic value expected Information Gain and the pragmatic value the preference or the constraint satisfies but again these are equivalent um or bounded derivations now in this figure here's pragmatic value here is epistemic value so just just briefly um and and these equations also again another area where where people can look is using the ontology here's natural language ways to read these equations because the letters don't matter it's about the ontological relationships so this is free energy expected free energy on P_i the policy prior distribution is an expectation about the variational distribution about the expected sequence of observations y Tilda conditioned upon which action we're going to choose about How likely those observations are in light of our preferences that's pragmatic value How likely is the observations how are those observations going to be linked with our prior preferences on observations and then epistemic value component the double line is like the KL Divergence separator looking at these two sides they're pretty similar like they both have have QX Tilda conditioned on P_i so that's my variational distribution of what latent States I expect to unfold conditioned upon the action that I choose the difference is here we have what are going what is my variational Distribution on the observations that I expect to see conditioned upon the latent States here is the observations I expect to see also conditioned upon the action those are what those mean so this is saying if the action is not expected to change anything about the observations received then the epistemic value of the action is going to be low whereas if conditioning upon the action leads to a large difference then it's going to be high okay now you brought it up in the context of like wanting a certain regime of action specializing to get by dropping out one of these terms you get certain subtypes that's kind of the break the puzzle down the build the puzzle up is you will

find many thousands of Articles discussing these topics here's a way that we can see that those are special cases of a more generic optimization Target function so if we only have pragmatic value we get expected utility Theory basian decision Theory but what would be the issue with that well we wouldn't be able to have learning what is the reinforcement learning approach to that they try to bring learning into the reward function what if you drop pragmatic value then all observations are equally preferable so behavior is driven by just surprise and learning and that's like novelty search or pure intrinsic motivation so this is sort of revealing and clarifying that expected free energy is an integrated decision-making imperative that contains some blend of a pragmatic value and some blend of an epistemic value so whether we choose now to zero out some of those sections to get certain Behavior or we just want a new lens to look back on the last Decades of signal processing and control theory this is just a helpful way to see that okay how to inter how to interpret 1 three four five yeah K Divergence between What observations I policy and the true distribution of the observations conditioned on my preferences so this is like KL optimal control and that this you know is used as an objective function in some control theoretic setting it's yeah yeah and there's other there's other imperative functions that can be drawn for example like um Baron millage and others have clarified like well there's expected free energy and then there's free energy of the expected future but so you know there's some other imperatives that you could draw up it's not like this is the only way to do policy updating and active inference what what do you mean with that exactly in the sense that the the variation of fre energy is the way to go but in exp energy there's discussion about what the right way is I'll I'll like link this in in the chat this is from 2020 here's another future oriented imperative function that still is a free energy so still can be used for V variational gradient descent and this is just another another family of imperatives to use for policy updating and this only the case with expected free energy and the other variation fore energy there is uh the consensus of this is the thing to do um yes there are different ways to approximate variational free energy like this is with um nor AED and it's it's like okay it's a slightly different approximator so there's there there's some there are some flavors and choices it's not that there's some finished singular answer especially when you start getting into love of data sets there are some implementations that may work you know for discreet or continuous or different kinds of different other um features of the specific issue but again this is like why low road and high road is so important because the high road and the analytical directions there they're system agnostic their data set independent they're not talking about computer languages it's not the operational order about what comes before this and how does this

variable relate to that per se on the low road that is the whole discussion and then the generative model is the artifact that is entirely constructed to be viable under those analytical top- down descriptions what is the total discussion in the low road that contrasts with the high road the the kind of spe specific implementation or it's it's um I mean contrast or complement again that's what chapters 2 and three are about and then we meet back up in chapter 4 with a train station that can be reached from both roads because in a low road we're talking about building these unified basian cognitive models that deal with the inbound perceptual and the out bound action selection period that has much to do with cybernetics endtoend variational Auto encoders and so on top down we're not thinking about one specific computer implementation let alone like a data set or a computer language we're thinking about the analytical descriptions of things in their persistence and change and then the active inference generative model is something like a variational autoencoder asterisk something like that but it could be that that is compliant and describable through that higho channel but the models that we're going to look at in chapter 7 even and 8 in discret in continuous time those are the kinds of generative models described in chapter 4 which are reached out from chapters 2 and three just so for me to understand also the hyro delivers one true value of the variation for energy that is delivers one but it is like open and then there are different ways to approximated uh but the Hy does leave out space for what the exact expect free energy formulation should be I'm sure people have different responses to that I would say there isn't a singular true variational free energy for a system because that's a feature of a model so the variational free energy is like an observer approximating a variable so if we were talking about well does does the temperature in the room have a real L2 Norm a real sum of squares no different observers could make different linear models and they would yield different sums of squares different R squares different P values Etc you know is there a true Anova F statistic for the effect of this on that or are there just statistic who are making different Anova and Reporting their results so that's the instrumentalist angle which is people can calculate the vfv for a model but the map is not the ter territory so there might be a vfe for a map but a map isn't the territory so that vfe doesn't describe the territory because somebody else could have chosen a different variational family and then they would have gotten a vfe basically everybody has their own Q right everybody is their own Q that's what Q is I mean q is like the individual making you know a model of the world that's that's where the Q comes from that's the approximation it's it's me as an agent trying to understand and I I have this generative model that I make of the process right is that the idea exactly and so everywhere you see a q or everywhere it's talking about a

map this is like a sort of that's why um Toby St Clair SMI in his dissertation in in live stream 54 on the category Theory accounting for the active inference generative models he calls it the compositional cognitive cartography so this is like the science of the map making because it sets up the accounting structure for making Maps but then how coar grain should the map be what's the scope what's the angle what's the ortho project I mean there's so many questions even if we were just making a map of neuroimaging which again returns to SPM the statistical parametric mapping making maps of activation in brain regions or of inferred neural activation patterns from neuroimaging and then going from that map of activation to the posterior probability Maps so statistical transformations of those Maps but by the time you get to realizing that you're dealing with the probability transforms of neuroimaging data including all the steps of like registration and processing any statistic you get out the other side it's pretty easy to see that that's a feature of the modeling process though it has a causal relationship with the brain but it's not like there's just one P value nor is there one posterior nor is there one VF for a given experiment and that experiment was just one of many that could have been done okay not sure I'm going to copy some of these great comments in here let's go to the like again there's like so many fun things hidden in here and and what isn't there is for us to add and enjoy okay nice um I have to run it's it's a thank you last time uh this was great um I hope to join the next ones thanks for the all the links and the Koda I'll check those out y thank you okay all everyone okay yes thank you okay so nacho wrote I always wondered if there is an estimation of the actual number of levels layers present in different species I believe the number of layers determine the inference abilities of the animal is that right yeah in fact this is an interesting thing that um Carl friston will often say oh well like we make layers with six we make models with six layers because there's six layers in the cortex but I've always been a bit perplexed by that answer because each cortical layer though it has six histologically distinct strata is actually one unit of nesting from a cognitive basian perspective so how would you determine the number of layers well there might be a situation where um you can tease that out experimentally but how does it relate to structure learning learning you could have inference on how many layers should I make this model and that's kind of where things are at in terms of like neuros symbolic programming and procedural generation of active inference generative models is exploring model classes that include different layers and types of nesting then Ben added gray points and and there's several papers that have this sort of two-stroke engine which also corresponds a little bit to expectation maximization em optimization on structure okay any last um comments you're muted Charles I said Ray kerswell has a nice

book called creating Minds which speaks to the six layers in the cortex as well I think so he's in agreement with Carl yeah the six layers are um are simply anatomical even though there's like Su layers and and blah blah blah but the each each cortical stack is act here's the six layers right here each cortical column is one layer of a basian hierarchical model so even if there was only two anatomical layers or 35 anatomical layers per column you could have a linear concatenation of 75 columns so people focus on the six because it's clear that there's something repetitive and important happening with the cortical layers that can be seen histologically from like an evolutionary perspective and all these other things but there's no limit to their concatenation laterally and how many nested layers that could have but if you look at this from like a general a generalized coord perspective like you have the position then the velocity then the acceleration you keep on going up the derivatives you could look at real time series and find out how much variance is in these higher moments and after you get through a few moments there's not much to explain for most time series would you say that another way yeah if you want to explain or track the movement of a particle like a car in traffic or bitcoin price or your blood sugar just there's some particle that's following some sort of like really yeah I got it the first moments of the statistical distribution is its position right then there's its velocity like just the vector of its direction then the acceleration how is that velocity changing these are like derivatives called generalized coordinates for Real Time series you can ask what fraction of the variance is being explained as I add more generalized coordinates and this is an empirical question so it's you could construct one that has a lot of explanatory value at the 12th derivative but just empirically the explanatory value starts to get eaten up really fast and higher derivatives are overwhelmed by noise and they're slow so over shorter time scales the lower derivatives which is to say nested models that would have a few layers explain lots of the variance yeah I got how many layers and how much of the variance you need to explain over what time scale is 100% empirical when when you're when you're looking at the cortex horizontally rather than the six layers vertically how does recursion operate there I'm not a biologist but I'm just curious about how the how recursion operates that's a good question yeah there's a lot of ways you could get re-entry as recursion or reflection and reflexivity as recursion nacho yeah I was uh when I was asking the question I was also thinking not just from a neurological perspective but also from the perspective of psychology of perception for example if you if you have to if you're perceiving something and you say uh in any of the examples you said before as in a person that is deciding to smoke and everything's mixed up in in different layers different levels first you have to know that what you have in front of you is

a cigarette so that means that you have built uh from the input the sensory input you've built like a concept of what it is a cigarette which is made of different lines and points and that goes from the routine to the thalamus and then to the cortical layers then goes to to to the cortex so what I was uh asking is is if there is an estimate of the number of layers you need because in each layer you make a different abstraction you start by raw data raw uh points and lines and then those lines are connected and then you can see uh how a table or a bottle or a cigarette is made from these lines and I guess this is like a new layer from raw data to a concept and then as you go further up in the in this ladder you get more abstract Concepts you know what I mean I'm not sure if I'm oh yes like this is well studied with Friston and others and this is flowing through brain regions with a visual cortices so there's like Pro but there's yeah like the visual processing chain just to get from photons hitting the retina to just the acknowledgment of this is a given object a lot goes on there but cognitive modeling with the a matrix you might just say all right we have we have a noisiness or a noisy Matrix that just identifies it and that's kind of coarse graining over all of those like Edge detection and shape detection steps but if you were studying the Neuroscience of vision you might delve more into looking at these Visual cortices and the what and the where stream all these yeah I I teach these at University I'm a professor of these what I'm trying to do is to connect what we know from this perspective the classical studies of visual perception to to the inference model you know to to the number of layers you need in the model to explain this this kind of stuff yeah awesome I look forward to continuing to unpack that and chapter five okay okay is probably going to be a great one for you because it's going to focus on these three human neural systems and that'll kind of give a little bit of a Playbook with many many references to papers but doesn't look at the visual but there's a lot of other good visual papers cool okay can I ask you one last question okay go for it is top of mine I'm gonna bring it back to movies you know that's that's my focus is figuring out how movies work and uh there's there's a phenomenon in the movies you know because surprise is a big part of making a movie engage you you know but even if you're watching a bad movie there's a thing called closure where you'll you know you're watching a bad movie and yet you'll sit around to the end just to see how it turns out because it's like the movie the brain is building a model of the world that's going on in the movie The Cinematic Universe that you're experiencing and it kind of wants to know the final answer before it leaves unless it's super boring and so on is there something in the model that would that would address that Psych iCal phenomenon of the need for closure I'm going to give a similar answer which is really learn the textbook learn the grammar and then you will

construct many different hypotheses about how that could be like there could be a person who prefers I'm the kind of person who finishes movies or my culture expects me to finish movies right so then independent of whether that's their favorite color on the screen or the narrative is unfolding according to preferences they still would fulfill this other preference but then that would be where you design these different kinds of differentiating hypotheses with optimal experimental design to to get at that but there's a variety of ways that that independent of someone's visual hypothesis they could have narrative preference independent of narrative or genre there could be just behavioral expectations there's so many reasons because now we're talking about not just that even the narrative processing we're talking about someone's interplay between their own bodily Logistics and what they want to do with their own attention in their day and their uncertainties it's just like yeah I don't care if if um if John and Sally end up together at the end of this movie but I'm gonna watch it anyway but you know that's like well I'm just wondering if it is it a preference thing or is it a need for the is it a testing the model in terms of I built a model of how this movie is working I just want to see if the model Works don't don't close off one road with it is it a or b just say if it were a then the generative model could look like this if it were B the generative model could look like this now you'll have two okay how would I differentiate between if it were a and if it were B what else could you just keep on unfolding and that's what the structured inquiry starts to look like not going from a qualitative and just saying well is it this or that well that doesn't sound viable then just go with this one but you sketch them out and then you kind of build up your space of models that you're considering even Loosely but you can do this computationally depending on what you have available M and then well if were a what would that what would we expect to see in this data set versus B but if a and b give you the same result on the data set you're not going to be able to distinguish them so what data set would you need that's a good answer thank yeah it's it's not just about a Vibe check up front okay thank you okay thank you all till next time